

Retrospective of Female Digital Art Pioneers

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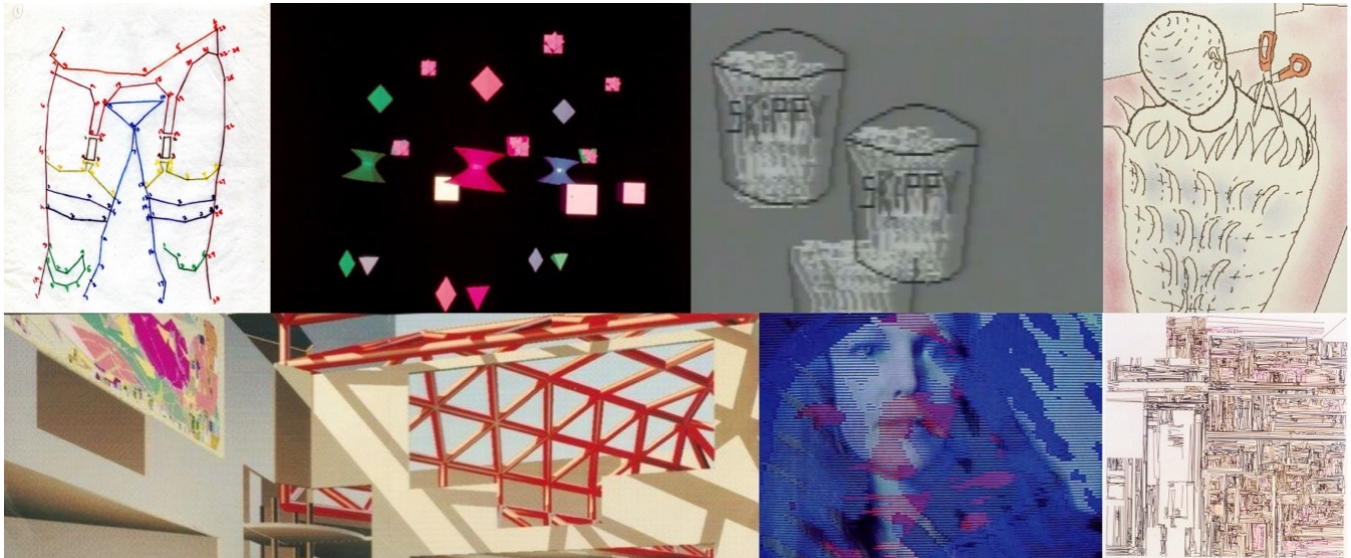
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1 ABOUT THE EXHIBITION

In celebrating the 50th year anniversary of SIGGRAPH, it is crucial to recognize and honor the contributions of female digital art pioneers by including them in the curated show as a retrospective. This decision not only acknowledges their groundbreaking work but also promotes gender equality, diversity, and inclusivity within the digital art community. The history of digital art has often overlooked or marginalized the achievements of women, despite their significant impact on the field. By featuring female digital art pioneers in a retrospective, SIGGRAPH can rectify this historical imbalance and provide a platform to showcase their innovative and influential creations. This acknowledgement is long overdue and can serve as an inspiration for future generations of artists, both male and female. Including female digital art pioneers in the curated show highlights their invaluable role in shaping the evolution of digital art over the past five decades. Their unique perspectives, artistic approaches,

and technical contributions have pushed boundaries, challenged norms, and enriched the creative landscape. Their participation not only adds depth and diversity to the exhibition but also offers a fresh lens through which to appreciate the multifaceted nature of digital art.

Moreover, showcasing female digital art pioneers in a retrospective aligns with the broader goals of promoting gender equality and inclusivity in the art world. By actively addressing the gender disparities that have persisted throughout history, SIGGRAPH demonstrates its commitment to creating an environment where all artists, regardless of gender, feel valued, recognized, and empowered. This sends a powerful message that diversity and representation are essential components of artistic excellence. Furthermore, featuring female digital art pioneers in the curated show serves as an educational opportunity for visitors and attendees. It allows them to learn about the achievements of these visionary artists, explore their unique artistic journeys, and gain a deeper understanding of the historical context in which their work was created. This knowledge fosters a more comprehensive appreciation of digital art's evolution and provides a foundation for further exploration and innovation.

In conclusion, including female digital art pioneers in the curated show as a retrospective not only supports the 50th year anniversary of SIGGRAPH but also promotes gender equality, diversity, and inclusivity in the digital art community. By recognizing and celebrating the contributions of these remarkable artists, SIGGRAPH paves the way for a more inclusive and vibrant future for digital art.

2 DESCRIPTION OF ARTWORK

This curated exhibition includes seven artists and eight artworks.

Girl Lifts Skirt (1974) A series of drawings were used to create one of the earliest computer animated artworks. In 1974, Allen used a computer and punch cards to create her first digital artworks. With an interest in inserting sensuality into the computer, this short animation of a woman seductively lifting her skirt served as a comment on the absence of the female perspective in the development of digital technology.

STEPS (1982) Inspired by Bauhaus Theater, this historical work was created as part of an art/research experiment merging performance and advanced technology, creating a dancing 3D human model and an illusion of dancing abstract forms. One of the earliest examples of human motion simulation created on the computer. Music by: Carter Burwell *Skippy Peanut Butter Jars* (1980) is an autobiographical story about how I program a computer to inspect my life. In the narrative I discuss and process my childhood beliefs that art is only important when it is old and includes naked female bodies. When I discovered the medium of the computer—where I could write a program to make an image—it was a revelation. I could be a multidimensional visual storyteller, using images, movement, words, and sound. So began my love affair with digital technologies or “computers”, as I would have said when I made this work in 1979-80. Technical Information: “*Skippy Peanut Butter Jars*” was created on one of the first Bally ZGrass 32 computers, which had a screen resolution of 160 x 100 pixels with 2 colors. I wrote the programs in the ZGrass language and then used a keyboard with a

joystick to make the drawings/animations. I stored them in array and saved the finished programs on audiotape. The piece ran in real time and was recorded on ¾-inch video, with two channels of audio added later.

Raffles City (1984), is a 3D computer animation, fully rendered, lighted and shaded. It's an animation of a flight around and into a 3D model of the I.M. Pei building under in Singapore. The duration of the animation is one minute in length. Sandy Pei welcomed Darcy Gerbarg to their office for the purpose of selecting which of their buildings she would use for her animated architectural art piece. Darcy Gerbarg selected Raffles City and worked with Ralph Heisel, lead architect at I.M. Pei. The animation, *Raffles City*, was created in 1984 at Magi Synthavision, by Darcy Gerbarg, Artist, and Eugene (Gene) Miller, VP Technology, Magi Synthavision.

Under Wraps (1989), is one of two similar digital images, created in 1989, using a Mac Plus computer. The complete artwork, consisting of two similar but not the same images, is intended to be seen as a 3 dimensional image. In order to create a 3 dimensional effect one needs to start with two identical images, side by side. Using the ‘lasso’ mode in the Mac Paint program, one could encapsulate an ‘object’ and then by moving it around, the image creates the illusion of moving forward or backward depending upon if it is moved to either the right or left side on the same plane. Finally, the two similar images need to be photographed and the 35mm slide film is then mounted between a 35mm slide paper holder and seen through a 3D viewer.

Grass Series V (1982) We have been thinking about composition for many years. In the context of picture making, ‘composition’ refers to the placement of a figure in the picture space. The type of picture or drawing makes no difference to the problem – ‘where shall the first mark go?’ Artists, even those working without benefit of computer, know that the very first mark is the key to what the work will become. Computer artists cannot just lay down the first brush stroke on a canvas wherever their intuition tells them. Some part of the program, some variable values, or some block of code represents the decision to place the first mark at a particular place. That part of the program must be thought out ahead of time. The placement of the first mark and the control of a drawing's white spaces are essential elements of the series of programs called GRASS. To help us solve our placement problem we used one of the standard ideas of recent computer graphics – divide and conquer, which also goes under the name ‘subdivision’. This strategy can be stated very simply – if the problem which faces you looks too hard, split it into two parts, store one of the parts, and work on the other. Chance are, the part of the problem you are now working will be easier to solve. If not, split this part into two, store one, and work on the other. Later, if necessary, you can figure out how to put the parts back together. ‘Divide and conquer’ works, when it works, because of limited resolution. If you start splitting a picture into pieces, you will finally get to a part of the picture which is the same size as the pen stroke. Going back to the original question of where to place the first mark in a drawing, we just admit that we don't know where it should go. The drawing rectangle is divided into two parts. Now we have two rectangles. Each of these is smaller than the first one. Put one of these rectangles aside and consider where to put the first mark in the other. This process is continued until we are working with a rectangle which is fairly small. Then we just draw a

simple line and start looking at the rectangles which were put aside. There are other decisions – how to split the rectangles in two and where to draw in the smallest – which we have not discussed, but this is the basic idea. The programs in the GRASS series have gone through an evolutionary process. The drawings you see were run on an Intertec SuperBrain, using a Watanabe plotter. The program is written in Microsoft BASIC.

On Becoming (1984) Expressing the conflicting layers of realities in our lives, as contexts transition, an abstracted digitized facial image is layered on a live video image captured in real time. Societal roles are inappropriately assigned creating frameworks to constrain our behavior. This image juxtaposes of an abstracted digitized image representing one aspect of my life, superimposed on live analogue video as I peer through the digital abstraction. Personal challenges and necessary transformations are expressed in *On Becoming*. The video digitizer, for me, was a sculpting process, creating images with light and time that embody powerful dynamics. The “self-image is a powerful communicator that injects humanness into the digital and creates a synthesis of woman and machine”. Through the fluidity created by the video digitizer, I created this layered composition. The photograph was created by photographing the image on the screen in real time. Tina Sauerlaender discusses in her article “A Short History of Self-Representation in Digital Art” for *Digital Art History Journal* in November 2020 © Springer-Verlag London Ltd., part of Springer Nature 2018 71 L. Candy et al., *Explorations in Art and Technology*, Springer Series.

Warpitout (1982) is an interactive piece that incorporates real-time morphing of an image of the players face. The player could use the controls on the unit to distort the image of themselves in real-time. It was exhibited at SIGGRAPH 1982 conference, in art museums, science museums, and other venues. A copy of *Warpitout* was purchased by the Ontario Science Center in 1984 as a permanent exhibit.

3 ARTIST INFORMATION

Rebecca Allen is an internationally recognized artist inspired by the aesthetics of motion, the study of perception and behavior and the potential of advanced technology. From the mid 1970's, Allen was a rare female artist working in the early stages of computer art and digital technology. Her pioneering artwork - which takes the form of experimental video, large-scale performances and virtual and augmented reality art installations - addresses the future of identity, nature, and what it means to be human as technology redefines our sense of reality. Rebecca moves fluidly between artist studio and research lab, using her research to inform her art. Though widely recognized as a contemporary visual arts medium, Allen was one of the first artists to utilize the computer as an artistic tool to make art involving human motion simulation, AI/artificial life algorithms and other generative techniques. Allen's work is exhibited internationally and is part of private and museum collections, including Whitney Museum, MoMA (NY), LACMA and Centre Georges Pompidou.

Copper Frances Giloth is an artist living between Amherst, MA, and Aix-en-Provence, France. She is Professor Emerita at the University of Massachusetts Amherst (UMass). Giloth's projects take

the form of drawings, books, animations, videos, websites, and installations. Her work has been exhibited in the United States, Japan, Canada, and across Europe. In February 2020, her most recent version of the VR installation “Labyrinth-of-Fables”—where visitors can experience the Versailles Labyrinth (constructed in 1665 and destroyed in 1775) by roaming in the garden and interacting with the 39-fable-inspired fountains—was first shown at Herter Gallery, UMass. Much of her earliest work was among the pioneering efforts in the then nascent field of computer art and computer graphics. She organized the international ACM SIGGRAPH Art competitions in 1982 and 1983. She is featured in *New Media Futures: The Rise of Women in the Digital Arts*, University of Illinois Press, 2018.

Darcy Gerbarg is an Abstract Expressionist painter with traditional Art training, “painting” AR and VR Artworks with colored light, in a 3DVR world, creating Interactive-Immersive Environments, digital paintings on canvas, paper, ceramics. She is a pioneer, first using Alvy Ray Smith's digital paint system at NYIT in 1979, and then other digital systems to make art in research labs where computer graphics, 3D animation, VR and AR were being explored and developed. Gerbarg has work in Art museums, galleries, fairs, corporate and private collections, internationally. Her 3D animation “Raffles City” was shown at SIGGRAPH and distributed in 1984. Her Immersive-Interactive VR Experience was exhibited at the CADAF Miami Art Fair 2019. Her Digital Paintings were featured at the Hampton's Virtual Art Fair 2020 and included in the physical Southampton Arts Center's, *Techspressionism: Digital and Beyond Exhibition 2022*. Darcy's Painting and Sculpture Garden - BRCvr2020-21 (Burning Man online) and VR Art Gallery Exhibition were active in AltspaceVR until that venue was shut down. Her work is shown: EADT Gallery, ArtGateVR; VR enhanced paintings on canvas are shown at PAKS International Art Exhibitions in Cannes, Basel, Paris, and at ASK, 2023. Gerbarg studied Art at the NY Studio School for Drawing, Painting and Sculpture, and in Woodstock. She has a long history of working with master printmakers, Krishna Reddy, Ronni Henning, and Kathy Caraccio. Gerbarg taught the first Computer Art course at NYU 1980, launched the first SIGGRAPH Art Shows 1981/1982, created the world's first MFA in Computer Art Program, at the School of Visual Arts, NYC mid-1980's, and was an adjunct professor in Computer Art, at SUNY Stony Brook and NYU.

Barbara Nessim (b. 1939, Bronx, NY) is an artist whose daring and prolific work, spanning five decades, defies narrow categorization. In its broadest sense, her artistic production has straddled fine art and illustration, all the while pushing against and reshaping the boundaries of the separation between the two fields. Nessim was one of the first artists in her profession to explore and make extensive use of the computer as an artistic tool at the dawn of the computer age in the early 1980s. Nessim's artworks have been exhibited at numerous institutions worldwide, including the Louvre, The Whitney Museum, The Cooper Hewitt Museum, The Smithsonian Institute and The Norman Rockwell Museum. Her work has graced the covers of *Time*, *Rolling Stone*, and *The New York Times Magazine* among others. Honors include the Pratt Lifetime Achievement Award (2015) and induction into the Society of Illustrators Hall of Fame (2020). Notable solo exhibitions include London's Victoria & Albert Museum (2013), Bard Graduate Center Gallery (2015), Malin Gallery (2021). Additionally, LA County Museum of Art (LACMA)

is to acquire three of Nessim's early digital artworks for their permanent collection and will be exhibited in their Retrospective of Female Digital Art Pioneers show in December, 2024.

Colette Stuebe Bangert (born 1934, Columbus, Ohio) is an American artist and new media artist who has created both computer-generated and traditional artworks. Her computer-generated artworks are the product of a decades-long collaboration with her husband, Charles Jeffries "Jeff" Bangert (1938–2019), a mathematician and computer graphics programmer. Bangert's work in traditional media includes painting, drawing, watercolor and textiles.

Joan Truckenbrod Intrigued with the potential of making ephemeral marks using a computer, I developed mathematical descriptions of natural phenomena, translated them into algorithms and incorporated into programs using a mainframe computer. I studied FORTRAN programming language and created a series of pen plotter drawings. Injecting a textural materiality into programmed imagery, I developed a process for creating coded algorithmic textiles. As a digital artist, I continue to propel the image on the screen into the material world, creating a physical presence. For me, textiles are intimately connected to computing. I am now working with the TC2 digital Jacquard Loom, creating hand-digital fiber art. This process is an interface of analogue and digital programming, combined with hand craft. A milestone group exhibit of early digital artworks currently at LACMA, Los Angeles County Museum of Art, included one of my early coded algorithmic textiles from 1979 titled "Electric Patchwork". This exhibit was titled Coded: Art

Enters the Computer Age 1952 - 1982, in the Spring of 2023. Four of my programmed plotter drawings created in 1975, and a coded algorithmic textile created in 1979, are in the Digital Art Collection at the Whitney Museum of American Art in New York where they were included in the exhibition Programmed: Rules, Codes and Choreographies in Art, 1965 - 2018, curated by Christiane Paul. Other collections including my digital artwork include the AKG Buffalo Museum, The Art Institute of Chicago and the Victoria and Albert Museum in London. My artwork is represented by RCM Galerie in Paris. Currently I am exploring connections between my hand-digital Jacquard weavings with 3D printing and laser embossing. I have a Courtesy Professorship at the University of Oregon and am working with the Center for Art Research Group and the Knight Science Center.

Jane Veeder is a pioneering video and digital media artist and educator who began working with electronic media in 1976. She is an American digital artist, filmmaker and educator. She was a professor at San Francisco State University in the Department of Design and Industry. Veeder is best known for her pioneering work in early computer graphics, however she has also worked extensively with traditional art forms such as painting, ceramics, theatre, and photography. In 1985, Veeder co-authored with Copper Giloith "The Paint Problem," an influential article in IEEE Computer Graphics and Applications on issues around the future of artists' digital tools. She has a long history of contributions to ACM SIGGRAPH as both a contributor and an organizer.